Department of CSE

B.Tech. CSE Trimester - 5

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Cloud Infrastructure and Services

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LAB RECORD

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Experiment - 3

AIM: To launch an Ubuntu EC2 instance, install the Nginx web server through a User Data script at launch time, and access the default page in a web browser. Record the differences compared to the Amazon Linux and Apache procedure.

Description

Amazon EC2 (Elastic Compute Cloud) provides virtual servers in the AWS cloud. In this experiment, an Ubuntu Server instance is launched and configured to install Nginx automatically during its first boot by using EC2 User Data. The instance security group allows HTTP traffic on port 80 so that the Nginx default webpage can be opened using the instance's public IPv4 address.

Prerequisites

• Active AWS account and access to the AWS Management Console.

• Basic understanding of EC2 instances and internet connectivity.

• A web browser.

• An existing EC2 key pair or the option to proceed without SSH access when only browser-based verification is required.

Steps

Part A: Launch an Ubuntu EC2 Instance

1. Log in to the AWS Management Console and open the EC2 Dashboard.

2. From the left navigation panel, select **Instances**, then click **Launch instances**.

3. Under Name and tags, enter an instance name such as **Ubuntu-Nginx-Server**.

4. Under Application and OS Images (AMI), select **Ubuntu** and choose an Ubuntu Server LTS AMI available in the region.

5. Select an eligible instance type, such as **t3.micro**, depending on availability and Free Tier eligibility.

6. Select an existing key pair or create a new key pair if SSH access is required.

7. Under Network settings, allow SSH traffic if required and allow HTTP traffic from the internet. HTTP uses port **80**.

8. Retain the default storage settings unless additional storage is required.

9. Expand Advanced details. Enter the User Data script shown in Part B, then launch the instance.

Part B: Install Nginx Using the User Data Script

In Advanced details, scroll to the User data text box and enter the following script. It updates Ubuntu package repositories, installs Nginx, enables the service at startup, and starts it.

```
#!/bin/bash

# Update Ubuntu package repositories
sudo apt update -y

# Install Nginx web server
sudo apt install nginx -y

# Enable Nginx to start automatically on boot
sudo systemctl enable nginx
```

```
# Start the Nginx web server
sudo systemctl start nginx
```

10. Verify that the script has been entered correctly in the User data field.

11. Click **Launch instance**.

12. Wait until the instance state becomes **Running** and status checks show **2/2 checks passed**.

13. Allow a few minutes for the User Data script to execute during the instance's first boot and complete the Nginx installation.

Note: Linux User Data scripts normally execute during the first boot only. If the script is changed after launch, it will not normally run again automatically.

Part C: Access the Nginx Default Web Page

14. From the EC2 Dashboard, select the newly launched Ubuntu-Nginx-Server instance.

15. Locate and copy its **Public IPv4 address**.

16. Open a web browser and enter **http://YOUR-PUBLIC-IP** in the address bar.

17. The Nginx default page should display the message **“Welcome to nginx!”**.

Ubuntu Terminal Verification (Optional)

If SSH access was configured, connect from a terminal using the downloaded key pair:

```
chmod 400 mykey.pem
ssh -i mykey.pem ubuntu@YOUR-PUBLIC-IP
sudo systemctl status nginx
curl http://localhost
```

The Nginx service should report **active (running)**. The curl command should return HTML for the default Nginx page.

Differences: Amazon Linux + Apache vs Ubuntu + Nginx

Feature	Amazon Linux + Apache	Ubuntu + Nginx
Operating system	Amazon Linux	Ubuntu Server
Package manager	yum / dnf	apt
Web server	Apache HTTP Server	Nginx
Install command	sudo yum install httpd -y (or dnf)	sudo apt install nginx -y
Service name	httpd	nginx
Start service	sudo systemctl start httpd	sudo systemctl start nginx
Enable at boot	sudo systemctl enable httpd	sudo systemctl enable nginx
Default HTTP port	80	80
Typical document root	/var/www/html	/var/www/html

Observations

• An Ubuntu EC2 instance was launched successfully through the AWS Management Console.

• The security group was configured to allow HTTP traffic on port 80.

• The User Data script automatically installed and started Nginx during the first boot.

• The Nginx default webpage was accessed through the instance's public IPv4 address.

• Ubuntu uses APT for package management, while Amazon Linux commonly uses YUM/DNF. The service names and installation commands differ between Apache and Nginx.

Output

Expected output: The browser displays the Nginx default welcome page with the heading **“Welcome to nginx!”**. The Nginx service is running on the Ubuntu EC2 instance and is accessible over HTTP using its public IPv4 address.

Result

Successfully launched an Ubuntu EC2 instance, installed and configured the Nginx web server automatically using an EC2 User Data script, and accessed the default Nginx webpage through a web browser.

Practical evidence to attach: Screenshot of the EC2 instance in Running state with 2/2 checks passed, the User Data script, and the Nginx Welcome page in the browser.

Actual Output Screenshot · Nginx Default Web Page

